

Real Analysis HW 6

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17th March 2026

BS - Page 104 Exercise 1

Determine a condition on $|x - 1|$ that will assure that

- $|x^2 - 1| < \frac{1}{2}$
- $|x^2 - 1| < \frac{1}{10^{-3}}$
- $|x^2 - 1| < \frac{1}{n}$ for a given $n \in \mathbb{N}$
- $|x^3 - 1| < \frac{1}{n}$ for a given $n \in \mathbb{N}$

If we want $|x^2 - 1| < \alpha$ for some $\alpha \in \mathbb{R}$, we need

$$|x + 1||x - 1| < \alpha$$

First we restrict $|x - 1| < 1 \Rightarrow |x + 1| < 3$

$$\Rightarrow |x - 1| < \min\{1, \frac{\alpha}{3}\}$$

Applying this to the first three parts, we get

- $|x - 1| < \min\{1, \frac{1}{6}\} = \frac{1}{6}$
- $|x - 1| < \min\{1, \frac{100}{3}\} = 1$
- $|x - 1| < \min\{1, \frac{1}{3n}\}$

For the last part, we need $|x^3 - 1| < \frac{1}{n}$, then

$$|x^3 - 1| = |x - 1||x^2 + x + 1| < \frac{1}{n}$$

Again if we restrict $|x - 1| < 1 \Rightarrow |x^2 + x + 1| < 4 + 2 + 1 = 7$, since $x \in (0, 2)$

$$|x - 1| < \min\{1, \frac{1}{7n}\}$$

BS - Page 104 Exercise 13

Let $c \in \mathbb{R}$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that $\lim_{x \rightarrow c} (f(x))^2 = L$

- Show that if $L = 0$, then $\lim_{x \rightarrow c} f(x) = 0$
- Show by example that if $L \neq 0$, then f may not have a limit at c

(a)

We have $\lim_{x \rightarrow c} (f(x))^2 = 0$, thus

$$\begin{aligned} \forall \epsilon > 0 \exists \delta \text{ such that } 0 < |x - c| < \delta \Rightarrow |f(x)^2| < \epsilon \\ \Rightarrow f(x)^2 < \epsilon \\ \Rightarrow |f(x)|^2 < \epsilon \end{aligned}$$

Also, $\sqrt{\epsilon} > 0, \Rightarrow |f(x)| < \sqrt{\epsilon}$

Thus $\lim_{x \rightarrow c} f(x) = 0$

(b)

$$\text{Let } f(x) = \begin{cases} -1, & x \leq c \\ 1, & x > c \end{cases}$$

Firstly, $f(x)^2 = 1 \forall x$, hence $\lim_{x \rightarrow c} (f(x))^2 = 1 \neq 0$

But,

Assume FTSOC, $\lim_{x \rightarrow c} f(x)$ exists

$$\Rightarrow \forall \epsilon > 0 \exists \delta \text{ such that } 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \epsilon$$

For $\epsilon = 0.1$ there exists δ_0 so that

If $x = c - \frac{\delta_0}{2}$, $f(x) = -1$ then

$$|-1 - L| < 0.1$$

If $x = c + \frac{\delta_0}{2}$, $f(x) = 1$ then

$$|1 - L| < 0.1$$

Clearly no such L exists, a contradiction

Thus $\lim_{x \rightarrow c} f(x)$ does not exist

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Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. Assume that $\lim_{x \rightarrow 0} f$ exists. Prove that $L = 0$ and then prove that f has a limit at every point $c \in \mathbb{R}$

(a)

Since $\lim_{x \rightarrow 0} f(x)$ exists and say is equal to L , we have

$$\forall \epsilon > 0 \exists \delta \text{ such that } 0 < |x| < \delta \Rightarrow |f(x) - L| < \epsilon$$

Also, we have the property that $f(x+y) = f(x) + f(y)$, thus $f(0) = f(x) + f(-x)$ then $-f(x) = f(-x)$

Then multiplying by -1 in our absolute value, we get,

$$|-f(x) + L| < \epsilon$$

$$|f(-x) + L| < \epsilon$$

Thus for $0 < |x| < \delta$, both of the inequalities hold

$$|f(-x) + L| < \epsilon, \quad |f(x) - L| < \epsilon$$

Letting $x = \frac{\delta}{2}$, we have $|f(-\frac{\delta}{2}) + L| < \epsilon$ and $|f(\frac{\delta}{2}) - L| < \epsilon$

Also letting $x = -\frac{\delta}{2}$, we have $|f(\frac{\delta}{2}) + L| < \epsilon$ and $|f(-\frac{\delta}{2}) - L| < \epsilon$

Thus, we see that $f(\frac{\delta}{2})$ is ϵ close to $-L$ and L , for this to hold it for all ϵ must be that $L = 0$

(b)

$$\begin{aligned} f(c) &= f(0+c) \\ &= f(0) + f(c) \\ &= 0 + f(c) \\ &= \lim_{\alpha \rightarrow 0} f(\alpha) + f(c) \\ &= \lim_{x \rightarrow c} f(x-c) + f(c) \\ &= \lim_{x \rightarrow c} [f(x-c) + f(c)] \\ &= \lim_{x \rightarrow c} f(x) \end{aligned}$$

Hence $\lim_{x \rightarrow c} f(x)$ does exist, given that $\lim_{x \rightarrow 0} f(x)$ exists

Moreover the limit is equal to $f(c)$

BS - Page 111 Exercise 14

Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$ and let $c \in \mathbb{R}$ be a cluster point of A . In addition, suppose that $f(x) \geq 0$, $\forall x \in A$ and let $\sqrt{f}$ be the function defined for $x \in A$ by $(\sqrt{f})(x) := \sqrt{f(x)}$. If $\lim_{x \rightarrow c} f$ exists, prove that $\lim_{x \rightarrow c} \sqrt{f} = \sqrt{\lim_{x \rightarrow c} f}$

$$L = \lim_{x \rightarrow c} f(x)$$

Case (i): $L = 0$

If $\epsilon > 0 \Rightarrow \epsilon^2 > 0$

If $L = 0$, we have $\forall \epsilon > 0 \exists \delta$ such that $0 < |x - c| < \delta \Rightarrow |f(x)| < \epsilon^2$

$$\Rightarrow \sqrt{|f(x)|} < \epsilon$$

$$\Rightarrow \sqrt{f(x)} < \epsilon$$

$$\lim_{x \rightarrow c} \sqrt{f(x)} = \sqrt{\lim_{x \rightarrow c} f(x)}$$

Case (ii): $L > 0$

If $\epsilon > 0 \Rightarrow \epsilon\sqrt{L} > 0$

Then, we have $\forall \epsilon > 0 \exists \delta$ such that $0 < |x - c| < \delta \Rightarrow |f(x) - L| < \epsilon\sqrt{L}$

$$\begin{aligned} |\sqrt{f(x)} - \sqrt{L}| &= \frac{|\sqrt{f(x)} - \sqrt{L}| |\sqrt{f(x)} + \sqrt{L}|}{|\sqrt{f(x)} + \sqrt{L}|} \\ &= \frac{|f(x) - L|}{\sqrt{f(x)} + \sqrt{L}} \\ &\leq \frac{|f(x) - L|}{\sqrt{L}} \\ &< \frac{\sqrt{L}\epsilon}{\sqrt{L}} \\ &< \epsilon \end{aligned}$$

Therefore $\forall \epsilon > 0 \exists \delta$ such that $0 < |x - c| < \delta \Rightarrow |\sqrt{f(x)} - \sqrt{L}| < \epsilon$

Thus $\lim_{x \rightarrow c} \sqrt{f(x)} = \sqrt{\lim_{x \rightarrow c} f(x)}$